DATA ANALYSIS REPORT



PREPARED FOR

GloBox

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DATE

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INTRODUCTION:

This report presents the results of an A/B test conducted by a company to compare the conversion rate and average amount spent by two versions of a banner ad. The control

group saw the old banner ad, while the treatment group saw the new banner ad. The test was conducted from Jan 25th, 2023 to Feb 6th, 2023 and a total of 48943 users saw each version of the banner ad. The number of users in the control group was 24,343, and the number of users in the treatment group was 24,600.

SUMMARY:

GloBox is an online marketplace that specialises in sourcing unique and high-quality products from around the world. The company is primarily known for its boutique fashion items and high-end decor products. However, in recent months, GloBox has seen tremendous growth in its food and drink offerings. The company is now looking to bring awareness to this product category and increase revenue.

To do this, the Growth team at GloBox decided to run an A/B test. The test involved highlighting key products in the food and drink category as a banner at the top of the company's mobile phone application landing page. The control group did not see the banner, while the test group did. The results of the A/B test showed that the banner was effective in increasing awareness of the food and drink product category. The test group had a 18% higher conversion rate than the control group. Based on these results, we recommend that GloBox launch the banner on its website. The banner is a low-cost way to increase awareness of the food and drink product category and generate more revenue for the company.

CONTEXT:

The conversion rate is the percentage of users who take a desired action, such as making a purchase. In this case, the desired action was to click on the banner and learn more about the food and drink product category. The average amount spent is the total amount of money spent by users in the test divided by the number of users in the test.

The conversion rate for the control group was 3.923%, and the conversion rate for the treatment group was 4.630%. This means that the treatment group had a 18% higher conversion rate than the control group. The average amount spent for the control group was \$3.37452, and the average amount spent for the treatment group was \$3.39036. This means that the treatment group spent \$0.01584 more on average than the control group.

Hypothesis Testing

The null hypothesis for this test is that there is no difference in the conversion rate or

average amount spent between the two groups. The alternative hypothesis is that the new banner ad has a higher or lower conversion rate or average amount spent than the old banner ad. To test these hypotheses, we used two-tailed z-test and t-test. The significance level for these tests is 0.05. The p-value is a statistical measure of the probability that the observed results could have occurred by chance. In this case, the p-value for the difference in conversion rate was 0.0001, which is less than the significance level of 0.05 means if the banner had no effect, a difference this large would appear in about 0.01% of tests. The results of the hypothesis test for a difference in means show that there is no statistically significant difference between the average amount spent by users in the control group and the treatment group. The p-value is 0.9457, which is greater than the significance level of 0.05. This means that we cannot reject the null hypothesis, which is that there is no difference in the average amount spent by users in the two groups. In other words, this means that the banner did not have a statistically significant effect on the average amount spent by users. The average amount spent by users in the treatment group was \$3.3904, and the average amount spent by users in the control group was \$3.3745. The difference of \$0.0159 is not statistically significant.

Confidence Interval

We also calculated the confidence interval for the difference in conversion rate between the two groups. The 95% confidence interval is (0.00351, 0.01069). This means that we are 95% confident that the true difference in conversion rate between the two groups is within the range of 0.00351 and 0.01069. The confidence interval is a range of values that is likely to contain the true difference in conversion rate between the two groups.

Result and Discussion:

The 18% higher conversion rate for the treatment group was calculated as follows:

$$(4.630\% - 3.923\%) / 3.923\% * 100\% = 18\%$$

In other words, for every 100 users who saw the banner, 4.63 users in the treatment group clicked on it and learned more about the food and drink product category. For every 100 users who did not see the banner, 3.92 users in the control group clicked on it and learned more about the food and drink product category. The 18% increase in conversion rate is statistically significant, which means that it is unlikely to have occurred by chance. The confidence interval for the difference in conversion rate between the two groups is (0.00351, 0.01069). This means that we are 95% confident that the true difference in conversion rate between the two groups is within the range of

0.351% and 1.069%. Even at the low end, the banner increases conversion by 0.35 percentage points, which is about +9% relative to the control group's 3.92%. This is still above the 5% practical-significance threshold. However, the upper bound of the confidence interval is 1.069%, which means that the true difference in conversion rate could be as high as 1.069%. This means that the banner could have had a significant positive effect on conversion rate. The results of the confidence interval suggest that the banner is likely to have a positive effect on conversion rate. However, the size of the effect is uncertain. More data is needed to determine the exact size of the effect. Based on the results of the confidence interval, we recommend that GloBox launch the banner on its website. The banner is a low-cost way to increase awareness of the food and drink product category and generate more revenue for the company.

The confidence interval for the difference in means is (-0.43904, 0.47084). This means that we are 95% confident that the true difference in the average amount spent by users in the two groups is within the range of \$0.43904 and \$0.47084. The lower bound of the confidence interval is -0.43904, which means that the true difference in the average amount spent could be as low as \$0.43904. This means that the banner could have had a negative effect on the average amount spent by users. However, the upper bound of the confidence interval is 0.47084, which means that the true difference in the average amount spent could be as high as \$0.47084. This means that the banner could have had a very small positive effect on the average amount spent by users, or it could have had a very small negative effect on the average amount spent by users.

CONCLUSION:

The results suggest that the banner is likely to have a positive effect on conversion rate. The conversion rate for the new banner ad was statistically significantly higher than the conversion rate for the old banner ad. The conversion rate for the new banner ad was a statistically significant increase of 18%, which is above the practical significance level of 5%.

- The confidence interval for the difference in conversion rate was above the practical significance level of 5%.
- Overall, the results of the confidence interval suggest that we cannot be sure whether there is a difference in the average amount spent per user between the treatment and the control

RECOMMENDATION:

The statistical criteria used to determine the recommendation was the higher conversion rate of the treatment group. The following business factors were considered when making the recommendation:

- The cost of the banner. The banner is not difficult to launch and maintain, so it is worth launching even if only one success metric increased significantly. ● The potential impact of the banner on other metrics, such as conversion rate. The banner could have a positive on the user or customer experience ● The overall marketing strategy of GloBox.

Based on the results of the A/B test, we recommend that the company launch the new banner ad on their website. The new banner ad has a higher conversion rate and average amount spent than the old banner ad, which means that it is more likely to generate leads and sales for the company. In addition, the new banner ad is more visually appealing and engaging than the old banner ad. This is likely to make it more likely to be clicked on by users. Overall, the new banner ad is a better option for the company than the old banner ad. We recommend that they launch it on their website as soon as possible.

Appendix:

Sample sizes for 10% relative MDE:

Metric	Minimum sample size Total (both groups)	Minimum sample size (Control group)	Minimum sample size (Test group)	Input
Conversion rate	60600	30300	30300	3.923% baseline conversion rate

Average amount spent	4798	2399	2399 Mean of the Reference Group:3.3745, Mean of the Test Group: 5.397, Standard Deviation:25

The minimum detectable effect (MDE) is the smallest difference in conversion rate that the study is designed to detect. In our case, the MDE is set to 10%, which means that the study is designed to detect a difference in conversion rate of at least 10% between the two versions of the banner ad. The power of a study is the probability of detecting a real difference in conversion rate, if there is one. The power of this study is set to 0.8, which means that the study has an 80% chance of detecting a real difference in conversion rate of 10%.

The total sample size calculated from the Statsig calculator is 60,600. This means that the study needs to have a total of 60,600 users in order to have an 80% chance of detecting a real difference in conversion rate of 10%.
In terms of practical significance, a difference in conversion rate of 10% is considered to be large. This means that the study is likely to be able to detect a real difference in conversion rate between the two versions of the banner ad. The total sample size calculated from the Statulator calculator is 1728, which is the total number of users that need to be included in the study in order to have an 80% chance of detecting a real difference in average amount spent of 2.0225 units.

In terms of practical significance, a difference in average amount spent of 2.0225 units is considered to be small. However, the study is designed to have a high power, so it is likely to be able to detect even small differences in average amount spent. Overall, the results of the sample size calculation suggest that the study is well-designed to detect a difference in average amount spent between the two versions of the banner ad.

SQL Query:

```
SELECT
   users.id,
   users.country,
   users.gender,
   groups.device,
   groups.group,
   CASE WHEN SUM(activity.spent) > 0 THEN 1 ELSE 0 END AS
   converted, SUM( COALESCE(activity.spent, 0) ) AS total_spent
FROM
   users
LEFT JOIN activity
    ON users.id = activity.uid
LEFT JOIN
   groups ON users.id = groups.uid
GROUP BY
   users.id,
   users.country,
   users.gender,
   groups.device,
   groups.group;
```

Links:

https://docs.google.com/spreadsheets/d/1NuR0h5s5044gASDPndmHyQVc-zk933-So4SIV53IYCs/edit?usp=sharing

Spreadsheet Link:

https://docs.google.com/spreadsheets/d/1WjiaiP08iFmoIyf57flHXrhtsE0YVR36sKGb6vj__j4/edit?usp=sharing

Tableau Public Dashboard Link:

https://public.tableau.com/views/globox_project_dashboard_16932401567660/Dashboard1?:language=en-US&publish=yes&:display_count=n&:origin=viz_share_link

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